

Actual sweep values

Each architecture is evaluated across the same three reservoir-control knobs.

Spectral radius

$$\rho(W^{res})$$

values

0.60 0.80 0.95 1.05

intuition

Recurrent gain:
larger values keep activity
circulating longer; near 1
tests the edge of stability.

Leak

$$\alpha$$

values

0.60 0.80 1.00

intuition

Update speed:
smaller leak mixes in the
new state more slowly;
 $\alpha = 1$ uses full update.

Input scaling

$$S_{in}$$

values

0.10 0.50 1.00 1.50

intuition

Input drive:
larger drive pushes richer
nonlinear response;
too large can saturate.

4 spectral radii x 3 leaks x 4 input scales = 48 settings per architecture / seed